

Machine Learning Foundations: Probability

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Use the terms and definitions below to understand concepts taught in the course.

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Term	Definition
Addition Rule	A probability rule stating that the probability of the union of two mutually exclusive events is the sum of their individual probabilities.
Bayes' Theorem	A mathematical formula used to determine the conditional probability of an event, given prior knowledge of related events.
Bernoulli Distribution	A discrete probability distribution for a random variable which has only two possible outcomes.
Binomial Distribution	A probability distribution that summarizes the likelihood that a value will take one of two independent values under a given set of parameters or assumptions.
Central Limit Theorem	A statistical theory that states, given a sufficiently large sample size, the sample mean of a random variable will be normally distributed.
Conditional Probability	The probability of an event occurring given that another event has already occurred.
Continuous Random Variable	A random variable that can take an infinite number of values within a given range.
Discrete Random Variable	A random variable that can take on a finite or countable number of possible values.
Experiment	A procedure carried out to support or refute a hypothesis, often used in the context of probability to describe a process that leads to one of several possible outcomes.
Event	A subset of the sample space, often defined to calculate probabilities of outcomes meeting certain criteria.
Likelihood	A concept related to probability, representing the strength of evidence in the data for a particular hypothesis.
Multiplication Rule	A probability rule that allows the calculation of the probability of the intersection of two independent events by multiplying their individual probabilities.
Naive Bayes Classifier	A simple probabilistic classifier based on applying Bayes' theorem with strong independence assumptions between the features.
Poisson Distribution	A probability distribution that describes the probability of a given number of events occurring in a fixed interval of time or space.
Probability	A numerical measure of the likelihood of a certain event, expressed as a number between 0 and 1.
Probability Distribution	A mathematical function that provides the probabilities of occurrence of different outcomes for a random variable.
Sample Space	The set of all possible outcomes of a random experiment.

